KAIZEN MATHS

Lesson starter

Confidence Intervals

Confidence Interval for a Mean

Name

S1

Calculate the confidence interval for a population mean using the known population standard deviation.

A sample of size 36 has mean 46.4. The population standard deviation is known to be 7.2. Find a 95% confidence interval for the population mean.

S2

Find the standard error and multiply by the critical value.

A population standard deviation is 4.8 and a sample size is 81. Find the margin of error for a 90% confidence interval for the mean.

S3

Interpret the confidence interval in context.

A 99% confidence interval for the mean waiting time is (79.76, 85.64) minutes. Interpret this interval.

S4

Calculate the confidence interval for a population mean using the known population standard deviation.

A sample of size 49 has mean 64.5. The population standard deviation is known to be 5.6. Find a 95% confidence interval for the population mean.

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Confidence Intervals

Confidence Interval for a Mean

Name

EA1

Find the standard error and multiply by the critical value.

A population standard deviation is 4.8 and a sample size is 49. Find the margin of error for a 95% confidence interval for the mean.

EA2

Interpret the confidence interval in context.

A 99% confidence interval for the mean waiting time is (79.07, 85.93) minutes. Interpret this interval.

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Confidence Intervals

Confidence Interval for a Mean

Name

EB1

Calculate the confidence interval for a population mean using the known population standard deviation.

A sample of size 49 has mean 75.5. The population standard deviation is known to be 7.2. Find a 95% confidence interval for the population mean.

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Confidence Intervals

Confidence Interval for a Mean

Name

EA1

Find the standard error and multiply by the critical value.

A population standard deviation is 4.8 and a sample size is 49. Find the margin of error for a 95% confidence interval for the mean.

EA2

Interpret the confidence interval in context.

A 99% confidence interval for the mean waiting time is (79.07, 85.93) minutes. Interpret this interval.

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Confidence Intervals

Confidence Interval for a Mean

Name

EB1

Calculate the confidence interval for a population mean using the known population standard deviation.

A sample of size 49 has mean 75.5. The population standard deviation is known to be 7.2. Find a 95% confidence interval for the population mean.

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Teacher answers & working

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S1

Calculate the confidence interval for a population mean using the known population standard deviation.

A sample of size 36 has mean 46.4. The population standard deviation is known to be 7.2. Find a 95% confidence interval for the population mean.

Answer:

95% CI: (44.048, 48.752).

Working:

Use $\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$.

For 95% confidence, z = 1.960.

Standard error = $\frac{7.2}{\sqrt{36}}$ = 1.2000.

Margin of error = 1.960 × 1.2000 = 2.3520.

Lower limit = 46.4 − 2.3520 = 44.048.

Upper limit = 46.4 + 2.3520 = 48.752.

So the 95% confidence interval is (44.048, 48.752).

S2

Find the standard error and multiply by the critical value.

A population standard deviation is 4.8 and a sample size is 81. Find the margin of error for a 90% confidence interval for the mean.

Answer:

Margin of error = 0.877.

Working:

Margin of error = $z\frac{\sigma}{\sqrt{n}}$.

For 90% confidence, z = 1.645.

Standard error = $\frac{4.8}{\sqrt{81}}$ = 0.5333.

Margin of error = 1.645 × 0.5333 = 0.877.

S3

Interpret the confidence interval in context.

A 99% confidence interval for the mean waiting time is (79.76, 85.64) minutes. Interpret this interval.

Answer:

We are 99% confident that the population mean waiting time lies between 79.76 and 85.64 minutes.

Working:

The interval estimates the population mean, not the sample mean.

The lower plausible value is 79.76 minutes.

The upper plausible value is 85.64 minutes.

Interpretation: We are 99% confident that the population mean waiting time lies between 79.76 and 85.64 minutes.

Teacher answers & working

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S4

Calculate the confidence interval for a population mean using the known population standard deviation.

A sample of size 49 has mean 64.5. The population standard deviation is known to be 5.6. Find a 95% confidence interval for the population mean.

Answer:

95% CI: (62.932, 66.068).

Working:

Use $\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$.

For 95% confidence, z = 1.960.

Standard error = $\frac{5.6}{\sqrt{49}}$ = 0.8000.

Margin of error = 1.960 × 0.8000 = 1.5680.

Lower limit = 64.5 − 1.5680 = 62.932.

Upper limit = 64.5 + 1.5680 = 66.068.

So the 95% confidence interval is (62.932, 66.068).

EA1

Find the standard error and multiply by the critical value.

A population standard deviation is 4.8 and a sample size is 49. Find the margin of error for a 95% confidence interval for the mean.

Answer:

Margin of error = 1.344.

Working:

Margin of error = $z\frac{\sigma}{\sqrt{n}}$.

For 95% confidence, z = 1.960.

Standard error = $\frac{4.8}{\sqrt{49}}$ = 0.6857.

Margin of error = 1.960 × 0.6857 = 1.344.

EA2

Interpret the confidence interval in context.

A 99% confidence interval for the mean waiting time is (79.07, 85.93) minutes. Interpret this interval.

Answer:

We are 99% confident that the population mean waiting time lies between 79.07 and 85.93 minutes.

Working:

The interval estimates the population mean, not the sample mean.

The lower plausible value is 79.07 minutes.

The upper plausible value is 85.93 minutes.

Interpretation: We are 99% confident that the population mean waiting time lies between 79.07 and 85.93 minutes.

Teacher answers & working

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EB1

Calculate the confidence interval for a population mean using the known population standard deviation.

A sample of size 49 has mean 75.5. The population standard deviation is known to be 7.2. Find a 95% confidence interval for the population mean.

Answer:

95% CI: (73.484, 77.516).

Working:

Use $\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$.

For 95% confidence, z = 1.960.

Standard error = $\frac{7.2}{\sqrt{49}}$ = 1.0286.

Margin of error = 1.960 × 1.0286 = 2.0160.

Lower limit = 75.5 − 2.0160 = 73.484.

Upper limit = 75.5 + 2.0160 = 77.516.

So the 95% confidence interval is (73.484, 77.516).